

Uputstvo za A4 OSPF protokol:

1. Povezivnje bežičnog rutera na Internet možemo uraditi preko kablovskog ili DSL modema. Vidimo da u

oblaku imamo samo modemski port i Eth6 ka Internetu. Biramo DSL modem  i povezujemo WiFi ruter na oblak.

U oblaku mora biti povezan Modem i Eth.

DSL

Modem5 <-> Ethernet6

From Port	To Port
Modem5	Ethernet6



2. Dodati u mrežu LAN bežični ruter i povezati ga na modem. Konfiguriramo Internet interfejs bežičnog rutera prema zahtevu zadatka i Portu Fa1/1 Ruteru1:

Router1

Physical Config CLI

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

FastEthernet1/1

FastEthernet1/1

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☒ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 000A.41EC.2C02

IP Configuration

IP Address 7.10.4.100

Subnet Mask 255.255.255.128

Tx Ring Limit 10

Wireless Router0

Physical Config GUI

GLOBAL

Settings

Algorithm Settings

INTERFACE

Internet

LAN

Wireless

Internet Settings

IP Configuration

☐ DHCP

☒ Static

☐ PPPoE

UserName

Password

Default Gateway 7.10.4.100

IP Address 7.10.4.102

Subnet Mask 255.255.255.128

DNS Server 13.103.177.12

3. LAN konfiguracija bežičnog rutera može da ostane po default-u:

LAN Settings

IP Configuration

IP Address 192.168.0.1

Subnet Mask 255.255.255.0

GUI bežičnog rutera – Internet Setup i Network Setup - treba da bude konfigurisano :

Wireless Router0

Physical Config GUI

Internet Setup

Internet Connection type Static IP

Internet IP Address: 7 . 10 . 4 . 102

Subnet Mask: 255 . 255 . 255 . 128

Default Gateway: 7 . 10 . 4 . 100

DNS 1: 13 . 103 . 177 . 12

DNS 2 (Optional): 0 . 0 . 0 . 0

DNS 3 (Optional): 0 . 0 . 0 . 0

Optional Settings (required by some internet service providers)

Host Name:

Domain Name:

MTU: Size: 1500

Network Setup

Router IP

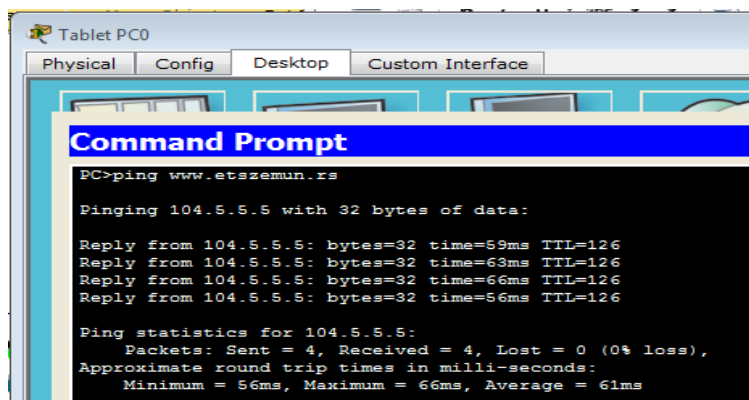
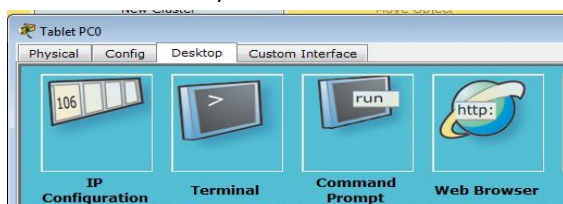
IP Address: 192 . 168 . 0 . 1

Subnet Mask: 255.255.255.0

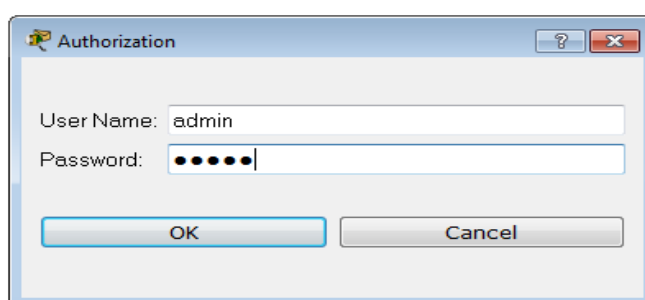
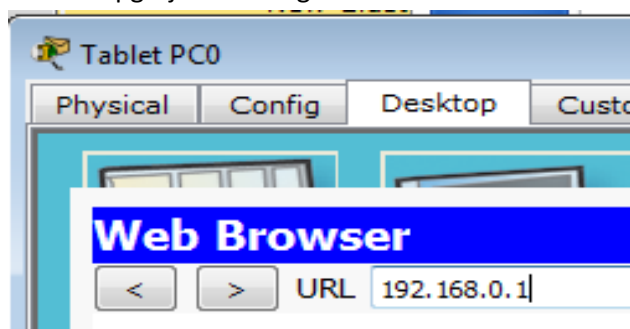
4. Dodajte bežičnog klijenta, npr. Tablet.

Trebalo bi da se automatski poveže, dobije IP adresu i izlazi na Internet

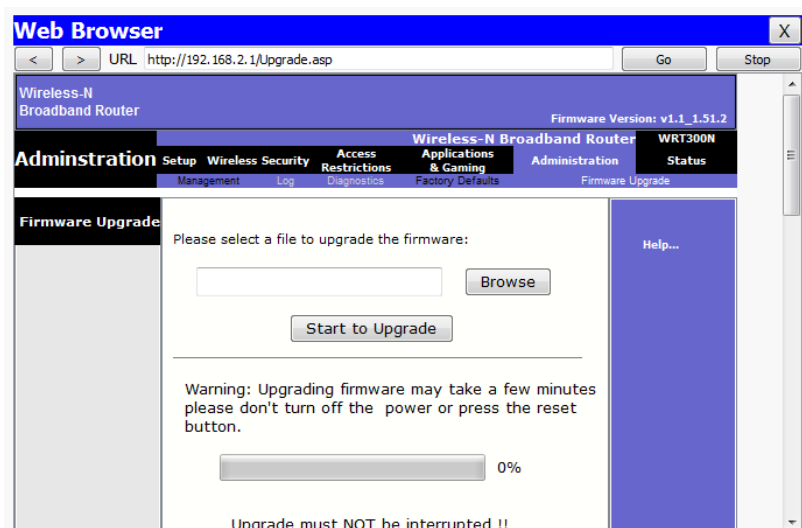
Može da se proveri sa tablet pomoću PINGa ili iz brauzera na WEB server: www.etszemun.rs (IP 104.5.5.5)



5. Apgrejd bežičnog rutera treba uraditi sa tableta kroz brauzer (nalog - admin/ admin):



Izabrati u meniju **Administration** pa **Upgrade Firmware**



Klikom na Browse dobijamo moguće verzije za firmware, odabrati NOVIJI i kliknuti na "Start to Upgrade". Dobijamo:



6. Prelazimo na odvojenu mrežu koja povezuje rutere WAN linkovima i u kojoj rutiranje treba odraditi OSPF protokolom. Analiziramo konfiguracije rutera i zahteve zadatka i popunjavamo tabelu zauzetih i slobodnih podmreža mreže 10.0.0.0/8. (**Napomena:** Tabela je samo radi preglednosti i lakšeg razumevanja, nije zahtevano!) Imamo samo konfiguraciju rutera u Beogradu, **Fa** interfejs je konfigurisan, a dva serijska za veze ka Novom Sadu i Kragujevcu nisu. Podignut je OSPF protokol za rutiranje.

interface FastEthernet0/0 ip address 10.0.0.126 255.255.255.128	interface Serial0/0 no ip address interface Serial0/1 no ip address	router ospf 1 log-adjacency-changes network 10.0.0.0 0.0.0.127 area 0
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Adrese za NS-LAN i KG-LAN nisu navedene, ali je dat broj potrebnih hostova. To nam određuje veličinu podmreža:

Ruter	Početna adr	Završna adr	Mrežna maska	Mrežna maska	Broj hostova	DG za LAN
Ruter3 BG	10.0.0.0	10.0.0.127	255.255.255.128	/25		10.0.0.126
NS	10.0.1.0	10.0.1.255	255.255.255.0	/24	200	10.0.1.254
KG	10.0.0.128	10.0.0.255	255.255.255.128	/25	100	10.0.0.254
NS-KG	10.0.2.0	10.0.2.3	255.255.255.252	/30	2	
BG-NS	10.0.2.4	10.0.2.7	255.255.255.252	/30	2	
BG-KG	10.0.2.8	10.0.2.11	255.255.255.252	/30	2	

7. Dodati dva rutera koji moraju da imaju po 2 serijska i jedan Eth interfejs i povezati ih u trougao sa ruterom u Beogradu.

8. Konfigurisati ruter NoviSad:

<i>hostname Ruter1</i> <i>interface FastEthernet0/0</i> <i>ip address 10.0.1.254 255.255.255.0</i>	<i>interface Serial0/0</i> <i>ip address 10.0.2.1 255.255.255.252</i>	<i>interface Serial0/1</i> <i>ip address 10.0.2.5 255.255.255.252</i>
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9. Konfigurisati ruter Kragujevac

<i>hostname Ruter2</i> <i>interface FastEthernet0/0</i> <i>ip address 10.0.0.254 255.255.255.128</i>	<i>interface Serial0/0</i> <i>ip address 10.0.2.2 255.255.255.252</i>	<i>interface Serial0/1</i> <i>ip address 10.0.2.10 255.255.255.252</i>
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10. Konfigurisati ruter Beograd, WAN portove (HDLC) prema NovomSadu i Kragujevcu:

<i>interface Serial0/0</i> <i>ip address 10.0.2.9 255.255.255.252</i>	<i>interface Serial0/1</i> <i>ip address 10.0.2.6 255.255.255.252</i>
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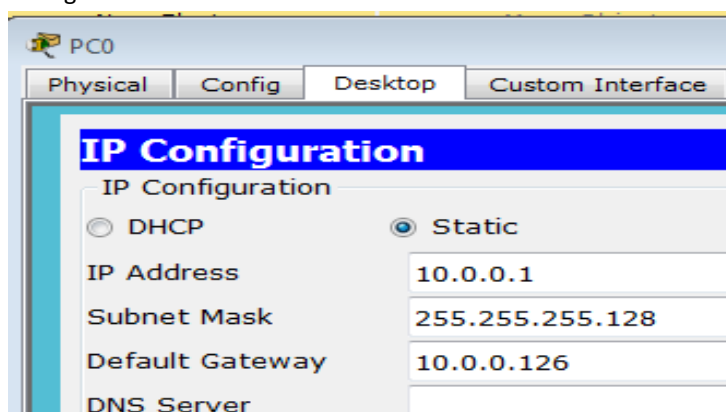
11. Proveriti veze između rutera komandom PING.

<i>Ruter1#ping 10.0.2.6</i> Type escape sequence to abort. Sending 5, 100-byte ICMP Echos to 10.0.2.6, timeout is 2 seconds: !!!! Success rate is 100 percent (5/5), round-trip min/avg/max = 1/3/7 ms	<i>Ruter1#ping 10.0.2.2</i> Type escape sequence to abort. Sending 5, 100-byte ICMP Echos to 10.0.2.2, timeout is 2 seconds: !!!! Success rate is 100 percent (5/5), round-trip min/avg/max = 1/8/21 ms
<i>Ruter3#ping 10.0.2.10</i> Type escape sequence to abort. Sending 5, 100-byte ICMP Echos to 10.0.2.10, timeout is 2 seconds: !!!! Success rate is 100 percent (5/5), round-trip min/avg/max = 5/9/21 ms	<i>Ruter3#ping 10.0.2.5</i> Type escape sequence to abort. Sending 5, 100-byte ICMP Echos to 10.0.2.5, timeout is 2 seconds: !!!! Success rate is 100 percent (5/5), round-trip min/avg/max = 1/9/17 ms

12. Konfigurisati OSPF ruting protocol na sva tri rutera:

Ruter1 – NoviSad: <i>router ospf 1</i> <i>log-adjacency-changes</i> <i>network 10.0.1.0 0.0.0.255 area 0</i> <i>network 10.0.2.0 0.0.0.3 area 0</i> <i>network 10.0.2.4 0.0.0.3 area 0</i>	Ruter2 – Kragujevac: <i>router ospf 1</i> <i>log-adjacency-changes</i> <i>network 10.0.0.128 0.0.0.127 area 0</i> <i>network 10.0.2.0 0.0.0.3 area 0</i> <i>network 10.0.2.8 0.0.0.3 area 0</i>	Ruter3 – Beograd: <i>router ospf 1</i> <i>log-adjacency-changes</i> <i>network 10.0.0.0 0.0.0.127 area 0</i> <i>network 10.0.2.4 0.0.0.3 area 0</i> <i>network 10.0.2.8 0.0.0.3 area 0</i>
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13. Konfigurisati statičku adresu za računar PC0:



14. Proveriti ruting tabele na ruterima:

```
Ruter1>
Ruter1>ena
Ruter1#sh ip route
Gateway of last resort is not set

    | 10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O    10.0.0.0/25 [110/65] via 10.0.2.6, 00:19:44, Serial0/1
C    10.0.2.0/30 is directly connected, Serial0/0
C    10.0.2.4/30 is directly connected, Serial0/1
O    10.0.2.8/30 [110/128] via 10.0.2.2, 00:18:44, Serial0/0
      [110/128] via 10.0.2.6, 00:18:44, Serial0/1

Ruter2>ena
Ruter2#sh ip route
Gateway of last resort is not set

    | 10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O    10.0.0.0/25 [110/65] via 10.0.2.10, 00:19:42, Serial0/1
C    10.0.2.0/30 is directly connected, Serial0/0
O    10.0.2.4/30 [110/128] via 10.0.2.1, 00:19:42, Serial0/0
      [110/128] via 10.0.2.10, 00:19:42, Serial0/1
C    10.0.2.8/30 is directly connected, Serial0/1

Ruter3>ena
Ruter3#sh ip route
Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.0.0/25 is directly connected, FastEthernet0/0
O    10.0.2.0/30 [110/128] via 10.0.2.5, 00:21:29, Serial0/1
      [110/128] via 10.0.2.10, 00:21:29, Serial0/0
C    10.0.2.4/30 is directly connected, Serial0/1
C    10.0.2.8/30 is directly connected, Serial0/0
```

C- označava direktno povezane mreže na interfejsima rutera

O –su mreže čije putanje je ruter saznao preko OSPF protokola.

Kada se pojave route sa oznakom O, znači da ruteri međusobom razmenjuju podatke o rutama preko OSPF protokola.

15. Proveriti PING od računara PC0 do Ruter1 (10.0.2.5), Ruter2 (10.0.2.10) i Ruter3 (10.0.0.126). Trebalo bi da sve funkcioniše.

```
PC0
Physical Config Desktop Custom Interface

Command Prompt

PC>ping 10.0.2.10

Pinging 10.0.2.10 with 32 bytes of data:

Reply from 10.0.2.10: bytes=32 time=1ms TTL=254
Reply from 10.0.2.10: bytes=32 time=1ms TTL=254
Reply from 10.0.2.10: bytes=32 time=1ms TTL=254
Reply from 10.0.2.10: bytes=32 time=9ms TTL=254

Ping statistics for 10.0.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 9ms, Average = 3ms

PC>ping 10.0.2.5

Pinging 10.0.2.5 with 32 bytes of data:

Reply from 10.0.2.5: bytes=32 time=14ms TTL=254
Reply from 10.0.2.5: bytes=32 time=1ms TTL=254
Reply from 10.0.2.5: bytes=32 time=1ms TTL=254
Reply from 10.0.2.5: bytes=32 time=1ms TTL=254

Ping statistics for 10.0.2.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 14ms, Average = 4ms
```

16. Ispraviti na sviču u Beogradu adresu za default gateway. Treba da bude 10.0.0.126:

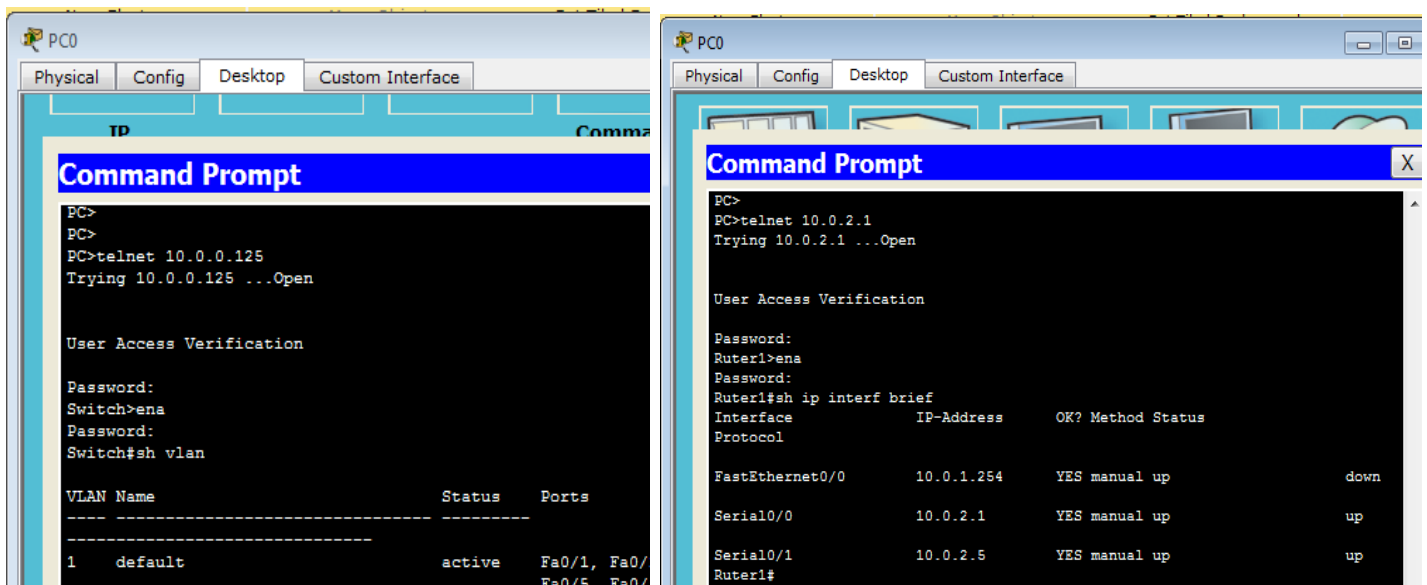
```
Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#ip default-gat 10.0.0.126
Switch(config)#exit
Switch#
```

17. Za daljinski pristup do Ruter1, Ruter2, Ruter3 i Swith - potrebno je konfigurisati (odnosno proveriti da li je već konfigurisan pristup preko TELNET-a):

Vidimo da nije konfigurisano <code>line con 0</code> <code>line aux 0</code> <code>line vty 0 4</code> <code>login</code>	Treba: Za Telnet <code>line vty 0 4</code> <code>password telnet</code> <code>login</code> Za konzolu <code>line con 0</code> <code>password console</code> <code>login</code>	Treba: <code>enable secret admin</code> <code>enable password cisco</code>
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18. Prijava na udaljeni uređaj:

Na primer sa PC0 na Svič i ruter Ruter1 (NoviSad)



19. Sačuvati konfiguracije svih rutera i svičeva (#**write**) i sačuvati na traženi direktorijum.

20. Popuniti obrasce: Прилог док2.doc , Прилог док3.doc i Прилог док4.doc